

Evaluation Report & Next-Step Execution Roadmap

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Assigned Construct: Attention

Proposed Task: Adaptive Shape Attention Task (ASAT)

Evaluation Status: Proceed with Modifications and Structured Modularization

1. Executive Summary

The proposed Adaptive Shape Attention Task (ASAT) demonstrates strong alignment with our vision of building a scientifically grounded, AI-driven cognitive assessment platform. The task is engaging, adaptive, difficult to fake, and capable of generating rich behavioral data. However, in its current form, ASAT measures a composite construct and may unintentionally capture Executive Functioning, Working Memory, Processing Speed, and Cognitive Flexibility in addition to Attention. To align with our startup objectives, the task should be redesigned into modular attention components while preserving adaptive and real-world characteristics.

2. Alignment with MentiScope's Objectives

- Alignment with Attention Theory: 9/10
- Scientific Validity: 8.5/10
- Innovation: 9/10
- Adaptive Potential: 9.5/10
- AI Readiness: 9.5/10
- MVP Feasibility: 9/10
- Overall Evaluation: 9/10 – Proceed with Modifications

3. Scientific Assessment

Primary attention components to measure:

- Sustained Attention
- Selective Attention
- Divided Attention
- Executive Attention

Potential construct contamination:

- Executive Functioning
- Working Memory (Gsm)
- Processing Speed (Gs)
- Cognitive Flexibility

The task should primarily measure attentional processes while minimizing unnecessary reasoning and memory demands.

4. Recommended Task Architecture

Module 1 – Sustained Attention

Rule remains fixed (e.g., click only when a blue triangle appears).

Measures: Vigilance, sustained attention, omission errors.

Module 2 – Selective Attention

Target appears among distractors.

Measures: Distractor resistance, focused attention, visual search efficiency.

Module 3 – Divided Attention

Students simultaneously monitor shapes, colors, and positions.

Measures: Attentional allocation and multi-source monitoring.

Module 4 – Executive Attention

Introduce controlled rule switching.

Measures: Conflict resolution, switching efficiency, executive control.

5. Recommended Scoring Model

- Sustained Attention Score – 25%

- Selective Attention Score – 25%

- Divided Attention Score – 20%

- Executive Attention Score – 30%

6. AI Analytics to Capture

- Average reaction time

- Reaction time variability

- Accuracy and error rate

- Switch cost

- Fatigue slope

- Adaptation speed

- Impulsivity index

- Attention stability index

- Recovery after errors

- Learning curve

7. MVP Deliverables (Current Phase)

- ✓ Procedural stimulus generator
- ✓ Rule-switch engine
- ✓ Adaptive difficulty engine
- ✓ Event logger
- ✓ Analytics dashboard
- ✓ Documentation
- ✓ Pilot dataset

8. Assessment SDK Compliance Requirements

All outputs must follow the common Assessment SDK specification.

Session Data:

student_id, session_id, construct, task_id, module_version, difficulty_level, start_time, end_time, timestamp.

Event Data:

event_id, student_id, session_id, construct, task_id, item_id, event_type, response, correct, reaction_time_ms, error_type, difficulty_level, timestamp.

Score Output:

raw_score, normalized_score, percentile, sub_scores, confidence_score.

Analytics Output:

behavioral_metrics and recommendations.

9. Next Sprint Execution Plan (2 Weeks)

Sprint Objective:

Redesign ASAT into scientifically distinct attention modules and develop the foundational MVP prototype.

Expected Deliverables:

1. Literature review on attention models and subdomains.
2. Construct map defining sustained, selective, divided, and executive attention.
3. Stimulus generator specifications (shapes, colors, positions, distractor density).
4. Rule-switch engine prototype.
5. Adaptive difficulty specifications (presentation speed, distractor count, switch frequency).

6. Event logging implementation.
7. Sample analytics dashboard.
8. Demonstration prototype of one attention module.

10. Definition of Done (DoD)

Scientific:

- ✓ Attention subdomains clearly defined.
- ✓ Construct contamination minimized.
- ✓ Difficulty levels calibrated.

Technical:

- ✓ Procedural item generator implemented.
- ✓ Rule-switch engine functional.
- ✓ Adaptive difficulty module implemented.
- ✓ Raw event logging and analytics dashboard available.
- ✓ Results exported in SDK format.

Validation:

- ✓ Test-retest reliability ≥ 0.80 .
- ✓ Internal consistency ≥ 0.80 .
- ✓ Pilot study with at least 100 students.
- ✓ Correlation studies with Working Memory and Processing Speed.

Guidance to the Intern

Your responsibility is not to build the entire platform at this stage. Your responsibility is to build a scientifically valid micro-assessment engine for Attention that follows the common Assessment SDK. Focus on creating modular attention assessments, behavioral logging, and adaptive difficulty mechanisms. This module will later be integrated into the unified cognitive and career assessment platform.